



DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: COMPILER DESIGN (24CS2235F)

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TITLE OF THE PROJECT	CodeFlow — Compiler-Based Code Flow Analyzer & Optimizer	
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Abstract

CodeFlow — Compiler-Based Code Flow Analyzer & Optimizer is a Compiler Design project developed to provide a practical and visual understanding of the major phases of a compiler. The system accepts a source program as input and processes it through multiple compiler phases, including lexical analysis, syntax analysis, intermediate code generation, code optimization, and code generation.

Initially, the lexical analyzer scans the source program and converts it into a sequence of tokens using regular expressions and finite automata concepts. These tokens are then processed by the syntax analyzer using grammar and parsing techniques to verify the syntactic structure of the program. After successful parsing, an Abstract Syntax Tree (AST) is constructed to represent the hierarchical structure of the source program.

The system then generates an intermediate representation such as Three-Address Code (TAC), which allows the program to be analyzed independently of a specific machine architecture. Basic blocks are identified and a Control Flow Graph (CFG) is generated to visually represent the flow of execution. The optimizer applies techniques such as constant folding, constant propagation, common subexpression elimination, dead-code elimination, and algebraic simplification to remove unnecessary computations and improve the intermediate code while preserving program behavior.

Finally, the optimized intermediate representation is converted into simplified target-machine instructions to demonstrate the code generation phase. The system provides visual representations of the transformations before and after optimization, allowing users to observe how the compiler processes and improves a program.

Overall, CodeFlow integrates important Compiler Design concepts into a single interactive system and provides a visual approach to understanding compiler phases, intermediate representations, control flow, and code optimization.

Signature of the Guide

HOD